AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T103/TM103
Computer Organization and Architecture

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 15/Apr./2026

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed:** 90 mins

Instructions:

- Total Marks: **30**
- This exam consists of **TWO** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **NOT** allowed.

PROTOTYPE

CamScanner

Part 1: Multiple Choices questions [5 Marks]

Q-1: Answer all the following questions by choosing the most correct statement.

1- Which computer level executes **machine language instructions directly?**
   a. Assembly Language level
   b. System Software level
   c. User Level
   d. Instruction Set Architecture level

2- Which base is commonly used internally by digital computers?
   a. Decimal
   b. Binary
   c. Hexadecimal
   d. Octal

3- 0.1 milliseconds are equivalent to:
   a. $1*10^{-2}$ microseconds
   b. $1*10^{-5}$ microseconds
   c. $1*10^{2}$ microseconds
   d. $0.1*2^{10}$ microseconds

4- Which signed number representation allows **two representations for zero**?
   a. Two's complement
   b. Signed magnitude
   c. One's complement
   d. Both choices b and c.

5- Which of the following is considered a **universal gate**?
   a. AND
   b. XOR
   c. NAND
   d. All the choices

PART 2: ALL QUESTIONS ARE REQUIRED [25 Marks]

Q-2: Given: **F(x, y, z) = (x'+z)(y+z).** Construct and complete the truth table, showing the intermediate results along with the output result. **[3 marks]**

Q-3: Represent **-19.5_{10}** in **14-bit floating-point (excess-16 bias)**. **[3 marks]**

Q-4: Represent the decimal number (**-110**) in binary using 8-bit Signed Magnitude, One's Complement, and Two's Complement: **[3 marks]**

Q-5: Given: **F(a, b, c) = ab + a'c + bc'** **[4 marks]**
Using De Morgan's law:
a) Complement each term.
b) Deduce the final expression for F'(a, b, c)

Q-6: Perform the following base conversions and show your work: **[4 marks]**
a) Convert **110101_2** to Decimal. [1 mark]
b) Convert **$2F_{16}$** to Decimal. [1 mark]
c) Convert **10101110.01_2** to Hexadecimal. [1 mark]
d) Convert **1010111.10_2** to Octal. [1 mark]

Q-7 : Given the SOP of function **F= xy'z' + x'yz' + x'yz + xyz': [8 marks]**
a. Draw the K-map of F, showing the groups that you have selected. (4 marks)
b. Write the simplified F. (2 marks)
c. Draw the logical Diagram of the simplified F. (2 marks)

End of questions

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